[Def. Let X be a Topological Space. A tuple of open sets $\{U,V\}$ is called Separation of X if

$$U \neq \emptyset, \quad V \neq \emptyset, \quad U \cap V = \emptyset, \quad U \cup V = X.$$

If the separation exists, then X is called disconnected.

[Thm. If X is connected and $f: X \to Y$ is continuous, then $f[X]$ is connected subspace.
Proof. Suppose that $f[X]$ is disconnected. Let $\{U,V\} \subseteq \mathcal{U}$ be a separation of $f[X]$.
That is, $U \neq \emptyset$, $V \neq \emptyset$, $U \cap V = \emptyset$, $U \cup V = f[X]$.
Then, $f^{-1}[U] \neq \emptyset$, $f^{-1}[V] \neq \emptyset$, $f^{-1}[U \cap V] = f^{-1}[U] \cap f^{-1}[V] = \emptyset$, $f^{-1}[U \cup V] = f^{-1}[U] \cup f^{-1}[V] = X$.

since f is continuous, $\{f^{-1}[U], f^{-1}[V]\}$ is a separation for X. contradiction.

[Thm.. If $A \subseteq X$ is connected and $A \subseteq B \subseteq \overline{A}$, then B is connected.
Proof. Suppose that B is disconnected. Then, there are opens $U, V \subseteq X$ such that

$$U \cap B \neq \emptyset, \quad V \cap B \neq \emptyset, \quad U \cap V \cap B = \emptyset, \quad U \cup V \supseteq B.$$

And, since $B \subseteq \overline{A}$, $U \cap \overline{A} \neq \emptyset$ and $V \cap \overline{A} \neq \emptyset$.
If $U \cap A = \emptyset$, then $A \subseteq U^c$, so $\overline{A} \subseteq U^c$. But this implies $\overline{A} \cap U^c = \emptyset$, contradiction.
Thus $U \cap A \neq \emptyset$, and similarly $V \cap A \neq \emptyset$. Moreover,

$$\emptyset = U \cap V \cap B \supseteq U \cap V \cap A, \text{ and } U \cup V \supseteq B \supseteq A, \text{ so}$$

$\{U, V\}$ is also separation of A, but it contradicts.

Thm. If subspaces $A_\alpha \subseteq X$ $(\alpha \in \Lambda)$ are connected and $\bigcap_{\alpha\in\Lambda} A_\alpha \neq \emptyset$, then $\bigcup_{\alpha\in\Lambda} A_\alpha$ is connected.

proof. Denote $A = \bigcup A_\alpha$. Suppose that A is disconnected by the separation $\{U, V\}$. That is, U, V are open s.t. $U \neq \emptyset$, $V \neq \emptyset$, $U \cap V = \emptyset$, $U \cup V = A$.
Let $a \in \bigcap_{\alpha\in\Lambda} A_\alpha$ be given. Then, for all $\alpha \in \Lambda$, $a \in A_\alpha$, thus $a \in A$.

If $a \in U$, put for each $\alpha \in \Lambda$, $U_\alpha = U \cap A_\alpha$ and $V_\alpha = V \cap A_\alpha$. Then, $a \in U \cap A_\alpha \neq \emptyset$, $U_\alpha \cap V_\alpha = U \cap V \cap A_\alpha = \emptyset$, $U_\alpha \cup V_\alpha = (U \cup V) \cap A_\alpha = A_\alpha$.
Since A_α is connected, $V_\alpha = V \cap A_\alpha$ must be empty. Hence,

$$V = V \cap A = V \cap \left(\bigcup_{\alpha\in\Lambda} A_\alpha\right) = \bigcup_{\alpha\in\Lambda} (V \cap A_\alpha) = \bigcup_{\alpha\in\Lambda} V_\alpha = \emptyset.$$

Contradiction. If $a \in V$, it can be proved similarly. ☒

Corollary. If $A_n \subseteq X$ $(n \in \mathbb{N})$ are connected, and for all $n \in \mathbb{N}$,

$$\left(\bigcup_{i=1}^{n} A_i\right) \cap A_{n+1} \neq \emptyset$$

then $\bigcup_{n=1}^{\infty} A_n$ is connected.

proof Take $B_n = A_1 \cup \cdots \cup A_n$. And using induction. (Use $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$)

Thm. Product of Connected space is Connected.

Proof. It holds in arbitrary product, but in this, prove any finite case

Let X, Y be Connected Space. For any $x \in X$ and $y \in Y$,

$$X \times \{y\} \cong X \quad \text{and} \quad \{x\} \times Y \cong Y$$

are connected. Moreover, $X \times \{y\} \cap \{x\} \times Y = \{(x,y)\} \neq \emptyset$. Therefore

$$(X \times \{y\}) \cup (\{x\} \times Y)$$

is connected. Let $b \in Y$ be fixed. Then, for any $x \in X$,

$$T_x = (X \times \{b\}) \cup (\{x\} \times Y)$$

is connected, and $T_x \supseteq X \times \{b\}$. Thus, $X \times \{b\} \subseteq \bigcap_{x \in X} T_x \neq \emptyset$.

Now, $X \times Y = \bigcup_{x \in X} T_x$ is connected.

$\dagger$ Arbitrary product case

Let X_α $(\alpha \in I)$ be Connected Spaces, and put $X = \prod_{\alpha \in I} X_\alpha$.

Let $(b_\alpha)_{\alpha \in I} \in X$ be fixed. Then, for any finite subset $\{\alpha_1, \cdots, \alpha_n\} \subseteq I$,

$$X_{(\alpha_1, \cdots, \alpha_n)} = \{ (x_\alpha)_{\alpha \in I} \in X \mid \alpha \notin \{\alpha_1, \cdots, \alpha_n\} \Rightarrow x_\alpha = b_\alpha \}.$$

Then, $X_{(\alpha_1, \cdots, \alpha_n)} \cong X_{\alpha_1} \times \cdots \times X_{\alpha_n}$, so it is Connected.

Moreover, for a Collection $\mathcal{F}$ of every finite subsets of I,

$$\bigcap_{\{\alpha_1, \cdots, \alpha_n\} \in \mathcal{F}} X_{(\alpha_1, \cdots, \alpha_n)} \quad \text{Contains } (b_\alpha), \text{ so non-empty,}$$

$$Y = \bigcup_{\{\alpha_1, \cdots, \alpha_n\} \in \mathcal{F}} X_{(\alpha_1, \cdots, \alpha_n)} \quad \text{is Connected.}$$

So, Y is also connected, enough to show $X = Y$.

Let $\prod_{\alpha \in I} U_\alpha \subseteq X$ be a basis element of X. By def, all but finitely many $U_\alpha = X_\alpha$.

So, for some finite $\{\alpha_1, \cdots, \alpha_n\} \subseteq I$ s.t $U_\alpha = X_\alpha$ if $\alpha \notin \{\alpha_1, \cdots, \alpha_n\}$

then $\prod_{\alpha \in I} U_\alpha \cap Y \neq \emptyset$. So proved

Def. Let X be a Topological space. Define a relation $\sim$ on X as:

$$x \sim y \iff \exists \text{ connected subspace } A \subseteq X \text{ s.t. } x \in A \text{ and } y \in A.$$

Then, it is equivalence relation; reflexivity and Symmetricity are clear, and transitivity: $x \sim y$ by A and $y \sim z$ by B, then $x \sim z$ by $A \cup B$.

Define Connected Component of X as:

$$C_x = [x]_\sim$$

Lemma. Let X be a Tps. Let $x \in X$ be given.

1). C_x is the larger Connected subset Containing x.

2). If $A \neq \emptyset$ is connected subset of X, then there is unique Connected Component $C \supseteq A$.

3). Connected Component is closed.

Proof of 1). Let $A \subseteq X$ be a Connected subset containing x. Then,

$$y \in A \Rightarrow x \sim y \Rightarrow y \in C_x, \text{ so } A \subseteq C_x.$$

Moreover, C_x is connected, because: given $z \in C_x$, there is a Connect Subset A_z s.t. $x, z \in A_z \subseteq C_x$. Now, since $\bigcap_{z \in C_x} A_z$ contains x, so nonempty, thus

$$C_x = \bigcup_{z \in C_x} A_z \text{ is connected.}$$

2) is trivial, Fixed $x \in A$, $a \in A \Rightarrow a \in C_x$. The uniqueness is given by the relativity

3). $C_x \subseteq \overline{C_x}$ trivially, and $\overline{C_x}$ is connected, so $\overline{C_x} \subseteq C_x$.

Thm. The Homeomorphism Preserves the Cardinality of Connected Component.

Proof. Let X and Y be Homeomorphic Topological Space with Homeomorphism $f: X \to Y$.
Given $x \in X$, there is a unique Connected Component of x, $C_x \subseteq X$.
Since the Connectedness is preserved by Continuous map, $f[C_x]$ is Connected subset, Containing $f(x)$. So, $f[C_x] \subseteq C_{f(x)}$. Similarly, $f^{-1}[C_{f(x)}] \subseteq C_{f^{-1}(f(x))} = C_x$.
Therefore, for any $x \in X$, $f[C_x] = C_{f(x)}$. Similarly, given $y \in Y$, $f^{-1}[C_y] = C_{f^{-1}(y)}$.

Now, define a map $\phi: \{C_x \mid x \in X\} \to \{C_y \mid y \in Y\} : C_x \mapsto C_{f(x)}$

Then, ϕ is one-to-one:

$C_{f(x_1)} = C_{f(x_2)} \Leftrightarrow f[C_{x_1}] = f[C_{x_2}]$, since f is Homeo, $C_{x_1} = C_{x_2}$

(Take f^{-1} to the both sides.) And, given C_y from the Codomain, there is $x \in X$ s.t. $y = f(x)$. So $\phi(C_x) = C_{f(x)} = C_y$. ☐

Thm. If a Topological Space X has only finite Connected Components, then every components are Clopen.

Proof. Let $C_1, \cdots, C_n$ be all connected components of X. Then, for each i,

$$C_i = X \setminus \left[\bigcup_{j \neq i} C_j \right]$$

is open.

Def. Let X be a Topological space.
We said X is locally connected at $x \in X$ if;
For any open $N \subseteq X$ containing $x \in X$, there is connected open V s.t $x \in V \subseteq N$.

X is called locally connected, if it is locally connected at every point.

Thm. X is locally Connected if and only if:
for any open $U \subseteq X$, every connected component of U is open.

Proof. Suppose that X is locally Connected space. Given an open set $U \subseteq X$, let $C \subseteq U$ be a Connected Component of the subspace U since X is l.c connected, given $x \in C$, there is a Connected open set V_x s.t $x \in V_x \subseteq U$.
Since C is Connected Component of x, $V_x \subseteq C$. So,

$$C = \bigcup_{x \in C} V_x \quad \text{is open set.}$$

Conversely, let $x \in N$ be given. Then, there is open Connected component $C_x \subseteq U$ exists.

Above definition and theorem are same in the case of Path-Connected.

Thm. If X is locally path-connected, then the connected component and path component are same.
Proof. Let $C \subseteq X$ be a Connected Component Given $x \in C$, let P_x be a path component of x.
Since P_x is path-Connected, so Connected, thus $P_x \subseteq C$.
To show $P_x = C$, suppose that $P_x \subsetneq C$. Let Q be the union of;

$$\{P_y \mid P_y \text{ is a path-component of } X \text{ s.t } P_y \neq P_x \text{ and } P_y \cap C \neq \emptyset\}.$$

Then, for each P_y is contained in C, so that $C = P_x \cup Q$.
Since X is locally path connected, P_x and Q are open, so these become the separation for C, it is contradiction.

Corollary. If X is locally path-connected, then
Path Connected $\Rightarrow$ Connected.

Thm. Quotient map preserves locally connected by image.

Proof. Let X be a locally connected and $p: X \to Y$ be a quotient map.

Let $C \subseteq Y$ be a connected component of an open set U of Y,

Then, clearly $p^{-1}[C] \subseteq p^{-1}[U]$. Given $x \in p^{-1}[C]$, there is a connected component $C_x \ni x$.

Then, $p(x) \in p[C_x]$ is connected, so $p[C_x] \subseteq C$, or $C_x \subseteq p^{-1}[C]$.

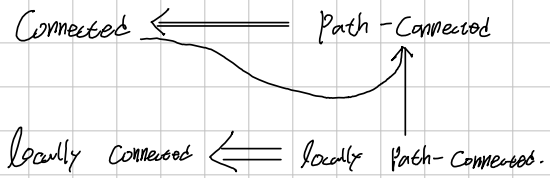
So, $p^{-1}[C] = \bigcup_{x \in p^{-1}[C]} C_x$ and since C_x are components of open set $p^{-1}[U]$,

So these are open, since X is locally connected.

Now, $p^{-1}[C]$ is open, so is C open, since p is quotient map. ☐

Thm. Quotient map preserves locally path-connected by image.

Summary.


Connected
Path - Connected
locally connected
locally path- connected.

Def. Let X be a Topological space.
If for any $a, b \in X$, there is a continuous map

$$p_a^b : [0,1] \to X \quad \text{s.t } p_a^b(0) = a \text{ and } p_a^b(1) = b$$

then X is called path-connected.

Thm. Path-Connected $\Rightarrow$ Connected.
Proof. Suppose that X is disconnected, but path-connected.
Let $\{U, V\}$ be the separation for X. Given $a \in U$ and $b \in V$,
there exists a Path $P : [0,1] \to X$ s.t $P(0) = a$, $P(1) = b$.
Since P is continuous and $[0,1]$ is connected, $\mathrm{Im}P$ is connected, but

$U \cap \mathrm{Im}P \ni a$, $V \cap \mathrm{Im}P \ni b$, $U \cap V \cap \mathrm{Im}P = \emptyset$, $U \cup V \supseteq \mathrm{Im}P$

So, $\{U, V\}$ is the separation for $\mathrm{Im}P$, it is Contradiction

Thm. Let X be a space. If $A_\alpha \subseteq X$ $(\alpha \in \Lambda)$ are path-connected with $\bigcap_{\alpha \in \Lambda} A_\alpha \neq \emptyset$,
then $\bigcup_{\alpha \in \Lambda} A_\alpha$ is path-connected.

Proof.

Topologist's Sine Curve.

Def. In $\mathbb{R}^2$, let $L = \{0\} \times [0,1]$ and $S = \{(x,y) \mid x > 0,\ y = \sin\frac{1}{x}\}$.

Then, the subspace $L \cup S$ is called the Topologist's Sine Curve.

It has the following properties:

1). Connected.

Since $x \mapsto \sin\frac{1}{x}$ $(x>0)$ is continuous, the graph S is connected.

And, since $L \cup S \subseteq \overline{S}$, $(L \subseteq S', \text{ so } L \cup S \subseteq S')$,

$L \cup S$ is also connected, by A connected $\Rightarrow$ $\forall B$ s.t. $A \subseteq B \subseteq \overline{A}$ is connected.

2. Not path-connected.

S is path-connected, since $(0,\infty)$ is path-connected and $x \mapsto \sin\frac{1}{x}$ is continuous.

But, for $(0,0) \in L$ and $(1, \sin 1) \in S$, there is no path between $(0,0)$ and $(1, \sin 1)$

(This is also the counter-example for S path. conn. but $\overline{S}$ is not.)

suppose that there is a path $p : [0,1] \to L \cup S$ from $(0,0)$ to $(1, \sin 1)$.

Put $t_0 = \sup\{t \in [0,1] \mid p(t) \in L\} = \sup p^{-1}[L]$.

Since L is compact, $t_0 \neq 1$. (If $t_0 = 1$, then $p(t)$ cannot be $(1, \sin 1)$.

Meanwhile, for any $\delta > 0$, there is $t_0 < t' < t_0 + \delta$ such that

$$\left| \pi_Y(p(t_0)) - \pi_Y(p(t')) \right| \geq \frac{1}{2}, \text{ so}$$

it is contradiction to p is continuous. (π_Y is a projection onto Y)